WORKED SOLUTION

STEP II 1992, Question 3

Pure • Hypocycloids and Area

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q3

The key idea

Everything follows from *rolling without slipping*: the length of arc the small circle has rolled along the big circle equals the length of arc of the small circle that has been in contact. That fixes how far the small circle has turned, and then B is just "centre P plus a unit vector in the right direction". For the sketch, it pays to notice that the coordinates simplify very neatly with the triple-angle formulae.

Solution

Finding the coordinates of B

Since the small circle (radius 1) stays in contact with the inside of the big one (radius 4), its centre is always at distance $4-1=3$ from O. With ϕ the angle AOP,

$$P = (3\cos\phi,\ 3\sin\phi).$$

Let T be the current point of contact. It lies on the ray OP produced, so $T = (4\cos\phi, 4\sin\phi)$ and the direction from P to T makes angle ϕ with the x-axis.

Initially B was at A, the point of contact. Since there is no slipping, the arc AT of the big circle has the same length as the arc BT of the small circle:

$$4\phi = 1 \times \angle BPT \quad \Longrightarrow \quad \angle BPT = 4\phi.$$

Ask yourself Is B at angle $\phi + 4\phi$ or $\phi - 4\phi$, measured at P?

As P moves anticlockwise around O, the small circle rolls along the inside of the big one and so turns *clockwise* (the arrow in the figure). The point B was the contact point earlier, so it has been left behind the new contact point T: going from T to B around the small circle is a clockwise turn of 4ϕ. Hence the direction $\overrightarrow{PB}$ makes angle $\phi - 4\phi = -3\phi$

with the x-axis, and since $PB = 1$,

$$\begin{aligned} B &= P + (\cos(-3\phi),\ \sin(-3\phi)) \\ &= (3\cos\phi + \cos 3\phi,\ 3\sin\phi - \sin 3\phi), \end{aligned}$$

as required. (Check: $\phi = 0$ gives $(4, 0) = A$.)

Sketching C

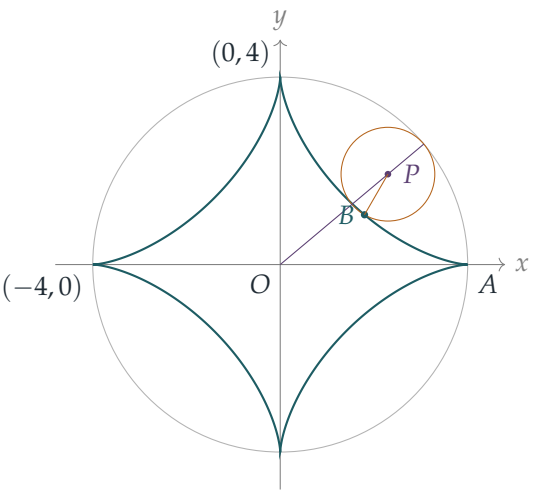
Using $\cos 3\phi = 4\cos^3\phi - 3\cos\phi$ and $\sin 3\phi = 3\sin\phi - 4\sin^3\phi$,

$$x = 4\cos^3\phi, \qquad y = 4\sin^3\phi.$$

So C is the curve $x^{2/3} + y^{2/3} = 4^{2/3}$, an *astroid*. Some features:

- B is on the big circle exactly when $\cos^6\phi + \sin^6\phi = 1$, i.e. $\phi = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$, giving the points $(\pm 4, 0)$, $(0, \pm 4)$. Here B is the contact point, so it is momentarily at rest and the curve has a cusp.
- The curve is symmetric in both axes (replace ϕ by $-\phi$ or $\pi - \phi$).
- In the first quadrant, $\dfrac{\mathrm{d}y}{\mathrm{d}x} = \dfrac{12\sin^2\phi\cos\phi}{-12\cos^2\phi\sin\phi} = -\tan\phi$, so the curve leaves $(4, 0)$ along the x-axis and arrives at $(0, 4)$ along the y-axis. It bows *inwards*: at $\phi = \frac{\pi}{4}$ it passes through $(\sqrt{2}, \sqrt{2})$, well inside the line $x + y = 4$.

After one circuit of P (ϕ from 0 to 2π) the small circle has turned 3 times relative to OP and B is back at A, so C is a closed curve with four cusps.



The area

We have $x = 3\cos\phi + \cos 3\phi$ and

$$\frac{\mathrm{d}y}{\mathrm{d}\phi} = 3\cos\phi - 3\cos 3\phi,$$

so

$$x\frac{\mathrm{d}y}{\mathrm{d}\phi} = 9\cos^2\phi - 6\cos\phi\cos 3\phi - 3\cos^2 3\phi.$$

Over a whole period, $\int_0^{2\pi}\cos^2 k\phi\,\mathrm{d}\phi = \pi$ for any integer $k \neq 0$ (write it as $\frac{1}{2}(1+\cos 2k\phi)$), and

$$\int_0^{2\pi}\cos\phi\cos 3\phi\,\mathrm{d}\phi = \frac{1}{2}\int_0^{2\pi}(\cos 4\phi + \cos 2\phi)\,\mathrm{d}\phi = 0.$$

Therefore

$$\text{Area} = \int_0^{2\pi} x\frac{\mathrm{d}y}{\mathrm{d}\phi}\,\mathrm{d}\phi = 9\pi - 0 - 3\pi = 6\pi.$$

The answer is positive, as it should be: B goes round C anticlockwise as ϕ increases.

Checks

- Since C bows inwards, the region lies inside the square with vertices at the four cusps, of area 32; and indeed $6\pi \approx 18.85 < 32$.
- Numerically, Simpson's rule on the integral and the shoelace area of a 200 000-gon inscribed in C both give $18.8496\ldots = 6\pi$.
- The standard astroid formula, area $\frac{3}{8}\pi a^2$ with $a = 4$, also gives 6π.
- At each cusp the numerical speed of B is zero, confirming that B is at rest when it touches the big circle, exactly as rolling demands.

Common slips

- Taking the angle of $\overrightarrow{PB}$ as $+5\phi$ or $+3\phi$: the small circle turns the *opposite* way to P's motion round O.
- Using the radius 4 instead of 3 for the path of P.
- Drawing rounded ends instead of sharp cusps, or arcs that bulge outwards.

Beyond STEP

Curves traced by a point on a circle rolling inside another are *hypocycloids*; rolling outside gives *epicycloids*. When the radii are in ratio 4 : 1 we get the astroid, and ratio 3 : 1 gives the three-cusped deltoid. The area formula $\int x\,\mathrm{d}y$ is a special case of Green's theorem, which relates an integral round a closed curve to a double integral over the region inside; it is a highlight of first-year vector calculus. Needham's *Visual Complex Analysis* treats these rolling curves elegantly using $z = 3e^{i\phi} + e^{-3i\phi}$.

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